1 A student is investigating the bending of a loaded wooden strip. Fig. 1.1 shows a rectangular strip of width *b* and thickness *t* overhanging the edge of a bench. A length *L* of the strip is unsupported.

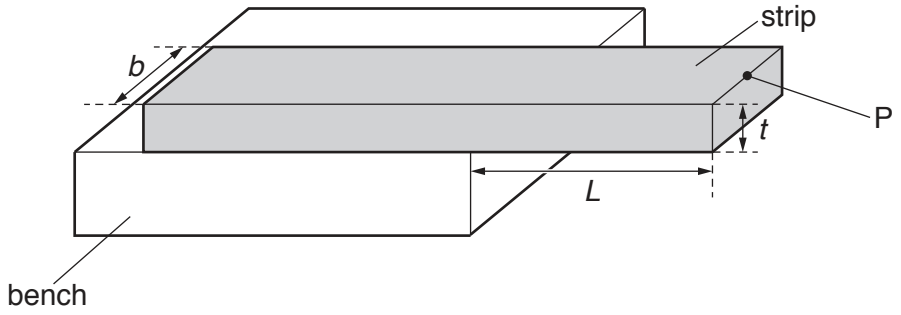


Fig. 1.1

A load of mass *M* is positioned at point P. This causes the unsupported part of the strip to bend with a deflection *s*, as shown in Fig. 1.2.

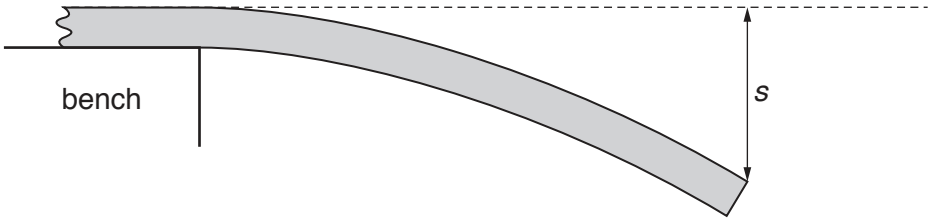


Fig. 1.2 (not to scale)

It is suggested that the relationship between *s* and *L* is

$$E = \frac{4MgL^3}{bst^3}$$

where *g* is the acceleration of free fall and *E* is the Young modulus of the wood.

Design a laboratory experiment to test the relationship between *s* and *L*.
Explain how your results could be used to determine a value for *E*.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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